B5: General Relativity — 2024 Model Solutions

Trinity Term 2024 (Paper A16729W1)

Question 1 — Perfect-fluid conservation, modified Einstein equations

Problem. (a) Find a current j^α obeying $j^\alpha{}_{;\alpha} = U^\alpha P_{,\alpha}$ from $T^{\alpha\beta}{}_{;\alpha} = 0$ for the perfect-fluid stress-energy. (b) Show that pressureless flow is geodesic. (c) The set $R_{\alpha\beta} - (1/n)Rg_{\alpha\beta} = 0$: count equations, find the special n, show R is constant. (d) Show that a trace-modified Einstein equation reduces to standard Einstein with cosmological constant. (e) Find Λ for $R_{\alpha\beta\mu\nu} = (K/12)(g_{\alpha\mu}g_{\beta\nu} - g_{\alpha\nu}g_{\beta\mu})$ in vacuum.

(a) Current from $T^{\alpha\beta}{}_{;\alpha} = 0$

Take the divergence of $T^{\alpha\beta} = Pg^{\alpha\beta} + (\rho + P/c^2)U^\alpha U^\beta$, using $g^{\alpha\beta}{}_{;\alpha} = 0$:

$$T^{\alpha\beta}{}_{;\alpha} = g^{\alpha\beta}P_{,\alpha} + \left[(\rho c^2 + P)U^\alpha U^\beta/c^2\right]_{;\alpha} = 0. \tag{1}$$

Define the candidate current:

$$j^\alpha \equiv (\rho c^2 + P)U^\alpha.$$

Insert into (1):

$$g^{\alpha\beta}P_{,\alpha} + \frac{1}{c^2}\left(j^\alpha U^\beta\right)_{;\alpha} = 0.$$

Expand the product:

$$\frac{1}{c^2}j^\alpha{}_{;\alpha}U^\beta + \frac{1}{c^2}j^\alpha U^\beta{}_{;\alpha} = -P^{,\beta}.$$

Contract with U_β, using $U_\beta U^\beta = -c^2$:

$$-j^\alpha{}_{;\alpha} + \frac{1}{c^2}j^\alpha U_\beta U^\beta{}_{;\alpha} = -U_\beta P^{,\beta}.$$

Apply $U_\beta U^\beta{}_{;\alpha} = \frac{1}{2}(U_\beta U^\beta)_{;\alpha} = 0$:

$$j^\alpha{}_{;\alpha} = U^\beta P_{,\beta}.$$

$$\boxed{j^\alpha = (\rho c^2 + P)U^\alpha, \qquad j^\alpha{}_{;\alpha} = U^\alpha P_{,\alpha}.} \tag{2}$$

(b) Pressureless flow is geodesic

Set $P = 0$ in $T^{\alpha\beta} = \rho U^\alpha U^\beta$. Take divergence:

$$(\rho U^\alpha)_{;\alpha}U^\beta + \rho U^\alpha U^\beta{}_{;\alpha} = 0.$$

Contract with U_β ($U_\beta U^\beta = -c^2$, $U_\beta U^\beta{}_{;\alpha} = 0$):

$$-c^2(\rho U^\alpha)_{;\alpha} = 0.$$

Hence rest-mass conservation:

$$(\rho U^\alpha)_{;\alpha} = 0.$$

Substitute back:

$$\rho U^\alpha U^\beta{}_{;\alpha} = 0.$$

$$\boxed{U^\alpha U^\beta{}_{;\alpha} = 0.} \tag{3}$$

(c) The set $R_{\alpha\beta} - (1/n)Rg_{\alpha\beta} = 0$

In 4D the symmetric tensor $R_{\alpha\beta}$ has $4(4+1)/2 = 10$ independent components, so the equations are 10. Take the trace $g^{\alpha\beta}(\cdot)$:

$$R - \tfrac{4}{n}R = 0 \implies \Big(1 - \tfrac{4}{n}\Big)R = 0.$$

For $n = 4$ the trace is automatically satisfied and is therefore not an independent equation:

$$\boxed{n = 4 \;\Rightarrow\; 9 \text{ independent equations.}} \tag{4}$$

Take the divergence of $R^\beta{}_\alpha = (R/n)\delta^\beta{}_\alpha$:

$$R^\beta{}_{\alpha;\beta} = \tfrac{1}{n}R_{,\alpha}.$$

Contracted Bianchi $R_{,\alpha} = 2R^\beta{}_{\alpha;\beta}$:

$$R_{,\alpha} = \tfrac{2}{n}R_{,\alpha} \implies \Big(1 - \tfrac{2}{n}\Big)R_{,\alpha} = 0.$$

With $n = 4$:

$$\tfrac{1}{2}R_{,\alpha} = 0 \implies \boxed{R = \text{const.}}$$

(d) Trace-free modified Einstein $\Rightarrow$ standard Einstein $+\Lambda$

Modified equation:

$$R_{\alpha\beta} - \tfrac{1}{4}Rg_{\alpha\beta} = -\tfrac{8\pi G}{c^4}\big(T_{\alpha\beta} - \tfrac{1}{4}Tg_{\alpha\beta}\big). \tag{5}$$

Take divergence; use Bianchi $\nabla^\beta R_{\alpha\beta} = \frac{1}{2}R_{,\alpha}$ and $T^{\alpha\beta}{}_{;\alpha} = 0$:

$$\tfrac{1}{2}R_{,\alpha} - \tfrac{1}{4}R_{,\alpha} = -\tfrac{8\pi G}{c^4}\Big(0 - \tfrac{1}{4}T_{,\alpha}\Big).$$

Simplify both sides:

$$\tfrac{1}{4}R_{,\alpha} = \tfrac{2\pi G}{c^4}T_{,\alpha}.$$

Integrate (constant of integration $\equiv 4\Lambda$):

$$R = \tfrac{8\pi G}{c^4}T + 4\Lambda. \tag{6}$$

Substitute (6) into (5):

$$R_{\alpha\beta} - \tfrac{1}{4}\big(\tfrac{8\pi G}{c^4}T + 4\Lambda\big)g_{\alpha\beta} = -\tfrac{8\pi G}{c^4}T_{\alpha\beta} + \tfrac{2\pi G}{c^4}Tg_{\alpha\beta}.$$

Collect T terms:

$$R_{\alpha\beta} = \Lambda g_{\alpha\beta} + \tfrac{4\pi G}{c^4}Tg_{\alpha\beta} - \tfrac{8\pi G}{c^4}T_{\alpha\beta}.$$

Use (6) to write $\frac{1}{2}Rg_{\alpha\beta} = \frac{4\pi G}{c^4}Tg_{\alpha\beta} + 2\Lambda g_{\alpha\beta}$:

$$R_{\alpha\beta} - \tfrac{1}{2}Rg_{\alpha\beta} = \Lambda g_{\alpha\beta} - 2\Lambda g_{\alpha\beta} - \tfrac{8\pi G}{c^4}T_{\alpha\beta}.$$

Rearrange:

$$\boxed{R_{\alpha\beta} - \tfrac{1}{2}Rg_{\alpha\beta} - \Lambda g_{\alpha\beta} = -\tfrac{8\pi G}{c^4}T_{\alpha\beta}.} \tag{7}$$

This is the standard Einstein equation with cosmological constant Λ.

(e) Maximally symmetric vacuum: Λ from K

Use the paper's convention $R_{\alpha\beta} \equiv -R^{\gamma}{}_{\alpha\beta\gamma}$. Raise the first index of the given Riemann tensor:

$$R^{\gamma}{}_{\alpha\beta\delta} = \tfrac{K}{12}\big(\delta^{\gamma}{}_{\beta} g_{\alpha\delta} - \delta^{\gamma}{}_{\delta} g_{\alpha\beta}\big).$$

Contract on $\gamma = \delta$:

$$-R^{\gamma}{}_{\alpha\beta\gamma} = -\tfrac{K}{12}\big(g_{\alpha\beta} - 4g_{\alpha\beta}\big).$$

Simplify:

$$R_{\alpha\beta} = \tfrac{K}{4} g_{\alpha\beta}.$$

Trace:

$$R = K.$$

Substitute into (7) with $T_{\alpha\beta} = 0$:

$$\tfrac{K}{4} g_{\alpha\beta} - \tfrac{K}{2} g_{\alpha\beta} - \Lambda g_{\alpha\beta} = 0.$$

Solve:

$$\boxed{\Lambda = -\tfrac{K}{4}.} \tag{8}$$

Question 2 — Reissner–Nordström geodesics

Problem. Charged BH metric, $A(r) = 1 - 2GM/(c^2 r) + GQ^2/(c^4 r^2)$. (a) Conserved k, h. (b) Effective-potential equation. (c) Particle from rest at infinity: turning point r_*. (d) Free fall from rest at r_+ when $r_Q > r_G$: $\dot r^2$ form, motion. (e) Proper time from r_+ to r_-.

(a) Conserved quantities along the geodesic

The metric is independent of t and ϕ. Lagrangian $2\mathcal{L} = g_{\mu\nu}\dot x^\mu \dot x^\nu$ with $\theta = \pi/2$, $\dot\theta = 0$. Euler–Lagrange in t:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\big(g_{tt}\dot t\big) = 0 \implies -Ac^2\dot t = \text{const.}$$

Define k by $A\dot t = k$:

$$\boxed{A\dot t = k.} \tag{9}$$

Euler–Lagrange in ϕ:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\big(r^2\sin^2\theta\,\dot\phi\big) = 0 \implies r^2\dot\phi = h.$$

$$\boxed{r^2\dot\phi = h.} \tag{10}$$

(b) Effective-potential equation

Normalisation $g_{\mu\nu}\dot x^\mu \dot x^\nu = -c^2$ on the equator:

$$-Ac^2\dot t^{\,2} + \frac{\dot r^{\,2}}{A} + r^2\dot\phi^{\,2} = -c^2.$$

Substitute (9),(10):

$$-\frac{c^2k^2}{A} + \frac{\dot r^{\,2}}{A} + \frac{h^2}{r^2} = -c^2.$$

Multiply by A:

$$\dot{r}^{\,2} = c^2k^2 - c^2A - \frac{Ah^2}{r^2}.$$

Halve and rearrange:

$$\tfrac{1}{2}\dot{r}^{\,2} + \tfrac{1}{2}c^2(A-1) + \frac{Ah^2}{2r^2} = \tfrac{1}{2}c^2(k^2-1).$$

Identify $V_{\text{eff}} = \frac{1}{2}c^2(A-1) + Ah^2/(2r^2)$. Insert $A = 1 - r_{\text{G}}/r + r_{\text{Q}}^2/r^2$, expand $A-1$:

$$\tfrac{1}{2}c^2(A-1) = -\frac{c^2r_{\text{G}}}{2r} + \frac{c^2r_{\text{Q}}^2}{2r^2}.$$

Expand $Ah^2/(2r^2)$:

$$\frac{Ah^2}{2r^2} = \frac{h^2}{2r^2} - \frac{h^2r_{\text{G}}}{2r^3} + \frac{h^2r_{\text{Q}}^2}{2r^4}.$$

Add:

$$\boxed{V_{\text{eff}}(r) = -\frac{c^2r_{\text{G}}}{2r} + \frac{c^2r_{\text{Q}}^2 + h^2}{2r^2} - \frac{h^2r_{\text{G}}}{2r^3} + \frac{(hr_{\text{Q}})^2}{2r^4}.} \tag{11}$$

(c) Stop point for a particle starting from rest at infinity

At rest at infinity: $h = 0$, $\dot{t} \to 1$, so $k = A(\infty)^{1/2}\dot{t} = 1$ and $k^2 - 1 = 0$:

$$\tfrac{1}{2}\dot{r}^{\,2} = -V_{\text{eff}}(r)\Big|_{h=0} = \frac{c^2r_{\text{G}}}{2r} - \frac{c^2r_{\text{Q}}^2}{2r^2}.$$

Set $\dot{r} = 0$:

$$\frac{r_{\text{G}}}{r} - \frac{r_{\text{Q}}^2}{r^2} = 0.$$

Solve for r:

$$\boxed{r_* = r_{\text{Q}}^2/r_{\text{G}}.} \tag{12}$$

At $r = r_*$ the radial acceleration $-V'_{\text{eff}}(r_*)$ is positive (charge term repulsive at small r): the particle bounces and accelerates back to infinity.

(d) Free fall from r_+ when $r_{\text{Q}} > r_{\text{G}}$

At rest at r_+ ($\dot{r} = 0$, $h = 0$): $k^2 = A(r_+)$. Plug into the radial equation with $h = 0$:

$$\dot{r}^{\,2} = c^2\big(k^2 - A(r)\big) = c^2\big(A(r_+) - A(r)\big).$$

Compute the difference:

$$A(r_+) - A(r) = r_{\text{G}}\Big(\frac{1}{r} - \frac{1}{r_+}\Big) - r_{\text{Q}}^2\Big(\frac{1}{r^2} - \frac{1}{r_+^2}\Big).$$

Reduce each fraction over the common denominator $r^2r_+^2$:

$$A(r_+) - A(r) = \frac{(r_+ - r)\big[r_{\text{G}}rr_+ - r_{\text{Q}}^2(r_+ + r)\big]}{r^2r_+^2}.$$

Factor the bracket as linear in r:

$$r_{\mathrm{G}} r r_+ - r_{\mathrm{Q}}^2 r_+ - r_{\mathrm{Q}}^2 r = (r_{\mathrm{G}} r_+ - r_{\mathrm{Q}}^2)\big(r - r_-\big), \qquad r_- \equiv \frac{r_{\mathrm{Q}}^2 r_+}{r_{\mathrm{G}} r_+ - r_{\mathrm{Q}}^2}.$$

Use $r_{\mathrm{Q}}^2 = r_{\mathrm{G}} r_*$ to rewrite the prefactor and the inner root:

$$r_{\mathrm{G}} r_+ - r_{\mathrm{Q}}^2 = r_{\mathrm{G}}(r_+ - r_*), \qquad r_- = \frac{r_+ r_*}{r_+ - r_*}.$$

Substitute:

$$\boxed{\dot{r}^2 = \frac{c^2 r_{\mathrm{G}}(r_+ - r_*)}{r_+^2\, r^2}\,(r_+ - r)(r - r_-).} \tag{13}$$

Both turning points are real with $r_- < r_+$. The particle oscillates between r_+ and r_-, never reaching $r = 0$: the charge term provides an inner centrifugal-like barrier. In Schwarzschild ($Q = 0$) one has $r_* = 0$, $r_- = 0$, so the inner barrier disappears and the particle reaches the singularity.

(e) Proper time $r_+ \to r_-$

Take the negative root of (13) (infall):

$$\mathrm{d}\tau = -\frac{r_+ r}{c\sqrt{r_{\mathrm{G}}(r_+ - r_*)}}\,\frac{\mathrm{d}r}{\sqrt{(r_+ - r)(r - r_-)}}.$$

Integrate from r_+ to r_-:

$$\tau = \frac{r_+}{c\sqrt{r_{\mathrm{G}}(r_+ - r_*)}}\int_{r_-}^{r_+}\frac{r\,\mathrm{d}r}{\sqrt{(r_+ - r)(r - r_-)}}.$$

Substitute $r = \frac{r_+ + r_-}{2} + \frac{r_+ - r_-}{2}u$, $\mathrm{d}r = \frac{r_+ - r_-}{2}\mathrm{d}u$:

$$(r_+ - r)(r - r_-) = \tfrac{(r_+ - r_-)^2}{4}(1 - u^2).$$

Reduce the integrand:

$$\frac{r\,\mathrm{d}r}{\sqrt{(r_+ - r)(r - r_-)}} = \frac{r}{\sqrt{1 - u^2}}\,\mathrm{d}u.$$

Split r and use parity (the u-piece integrates to zero):

$$\int_{-1}^{1}\frac{r}{\sqrt{1 - u^2}}\,\mathrm{d}u = \frac{r_+ + r_-}{2}\int_{-1}^{1}\frac{\mathrm{d}u}{\sqrt{1 - u^2}} = \frac{\pi(r_+ + r_-)}{2}.$$

Substitute:

$$\tau = \frac{\pi r_+(r_+ + r_-)}{2c\sqrt{r_{\mathrm{G}}(r_+ - r_*)}}.$$

Use the algebraic identity $r_+^2 = (r_+ - r_*)(r_+ + r_-)$ (from $r_- = r_+ r_*/(r_+ - r_*)$):

$$\frac{r_+}{\sqrt{r_{\mathrm{G}}(r_+ - r_*)}} = \sqrt{\frac{r_+ + r_-}{r_{\mathrm{G}}}}.$$

Insert:

$$\boxed{\tau = \frac{\pi r_{\mathrm{G}}}{2c}\Big(\frac{r_+ + r_-}{r_{\mathrm{G}}}\Big)^{3/2}.} \tag{14}$$

Question 3 — Anisotropic cosmology

Problem. Bianchi-I metric $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + e^{2\alpha}\mathrm{d}x^2 + e^{2\beta}(\mathrm{d}y^2 + \mathrm{d}z^2)$ with anisotropic perfect fluid (pressures $w_A\rho c^2$ along x, $w_B\rho c^2$ along y, z). (a) Christoffels. (b) $T^{\mu\nu}{}_{;\mu} = 0$ gives $\rho \propto e^{-(1+w_A)\alpha - 2(1+w_B)\beta}$. (c) Einstein equations. (d) $w_A = w_B = -1$: isotropic stable de Sitter. (e) Volume growth. (f) Two observational tests of cosmic anisotropy.

(a) Christoffel symbols

Use $x^0 = ct$, so $g_{00} = -1$ and $\partial_0 = (1/c)\partial_t$. Diagonal metric formula $\Gamma^\lambda_{\mu\nu} = \frac{1}{2}g^{\lambda\sigma}(g_{\sigma\mu,\nu} + g_{\sigma\nu,\mu} - g_{\mu\nu,\sigma})$ with $\sigma = \lambda$.

Compute Γ^0_{xx} ($\mu = \nu = x$, $\lambda = 0$):

$$\Gamma^0_{xx} = \tfrac{1}{2}g^{00}(-g_{xx,0}) = \tfrac{1}{2}(-1)(-\partial_0 e^{2\alpha}) = \frac{\dot\alpha\, e^{2\alpha}}{c}.$$

Same calculation for Γ^0_{yy}, Γ^0_{zz} with $g_{yy} = g_{zz} = e^{2\beta}$:

$$\Gamma^0_{yy} = \Gamma^0_{zz} = \frac{\dot\beta\, e^{2\beta}}{c}.$$

Compute Γ^x_{0x} ($\mu = 0, \nu = x, \lambda = x$):

$$\Gamma^x_{0x} = \tfrac{1}{2}g^{xx}g_{xx,0} = \tfrac{1}{2}e^{-2\alpha}\partial_0 e^{2\alpha} = \frac{\dot\alpha}{c}.$$

Same for y, z:

$$\Gamma^y_{0y} = \Gamma^z_{0z} = \frac{\dot\beta}{c}.$$

$$\boxed{\Gamma^0_{xx} = \frac{\dot\alpha}{c}e^{2\alpha}, \quad \Gamma^0_{yy} = \Gamma^0_{zz} = \frac{\dot\beta}{c}e^{2\beta}, \quad \Gamma^x_{0x} = \frac{\dot\alpha}{c}, \quad \Gamma^y_{0y} = \Gamma^z_{0z} = \frac{\dot\beta}{c}.} \tag{15}$$

(b) Continuity for ρ

The matrix in the question is $T^\mu{}_\nu = \mathrm{diag}(-\rho c^2, w_A\rho c^2, w_B\rho c^2, w_B\rho c^2)$. Then $T^{00} = g^{00}T^0{}_0 = \rho c^2$. Conservation $\nabla_\mu T^{\mu 0} = 0$ with $T^{i0} = 0$:

$$\partial_0 T^{00} + \Gamma^\mu_{\mu 0}T^{00} + \Gamma^0_{\mu\lambda}T^{\mu\lambda} = 0.$$

Insert $\Gamma^\mu_{\mu 0} = (\dot\alpha + 2\dot\beta)/c$:

$$c\dot\rho + \frac{\dot\alpha + 2\dot\beta}{c}\rho c^2 + \Gamma^0_{\mu\lambda}T^{\mu\lambda} = 0.$$

Compute $\Gamma^0_{\mu\lambda}T^{\mu\lambda} = \Gamma^0_{xx}T^{xx} + 2\Gamma^0_{yy}T^{yy}$, with $T^{xx} = w_A\rho c^2 e^{-2\alpha}$, $T^{yy} = w_B\rho c^2 e^{-2\beta}$:

$$\Gamma^0_{\mu\lambda}T^{\mu\lambda} = \frac{\dot\alpha w_A + 2\dot\beta w_B}{c}\rho c^2.$$

Combine and divide by $c\rho$:

$$\frac{\dot\rho}{\rho} = -(1 + w_A)\dot\alpha - 2(1 + w_B)\dot\beta.$$

Integrate:

$$\boxed{\rho(t) \propto e^{-(1+w_A)\alpha(t) - 2(1+w_B)\beta(t)}.} \tag{16}$$

(c) Einstein field equations

With $\ln|g| = 2\alpha + 4\beta$, the given Ricci formula becomes:

$$R_{\mu\nu} = \tfrac{1}{2}\partial_\mu\partial_\nu \ln|g| - \partial_\gamma \Gamma^\gamma_{\mu\nu} + \Gamma^\delta_{\mu\gamma}\Gamma^\gamma_{\nu\delta} - \tfrac{1}{2}\Gamma^\gamma_{\mu\nu}\partial_\gamma \ln|g|.$$

R_{00}. Only the first and third terms survive ($\Gamma^\gamma_{00} = 0$):

$$R_{00} = \tfrac{1}{2c^2}(2\ddot{\alpha} + 4\ddot{\beta}) + \tfrac{1}{c^2}(\dot{\alpha}^2 + 2\dot{\beta}^2).$$

Simplify:

$$R_{00} = \tfrac{1}{c^2}\big(\ddot{\alpha} + 2\ddot{\beta} + \dot{\alpha}^2 + 2\dot{\beta}^2\big).$$

R_{xx}. Only Γ^0_{xx} contributes:

$$\partial_0 \Gamma^0_{xx} = \frac{e^{2\alpha}}{c^2}(\ddot{\alpha} + 2\dot{\alpha}^2).$$

Quadratic term $\Gamma^\delta_{x\gamma}\Gamma^\gamma_{x\delta}$: the pairs $(\delta,\gamma) = (0,x),(x,0)$ contribute $(e^{2\alpha}\dot{\alpha}/c)(\dot{\alpha}/c)$ each:

$$\Gamma^\delta_{x\gamma}\Gamma^\gamma_{x\delta} = \frac{2e^{2\alpha}\dot{\alpha}^2}{c^2}.$$

Last term:

$$\tfrac{1}{2}\Gamma^0_{xx}\partial_0 \ln|g| = \tfrac{1}{2}\frac{e^{2\alpha}\dot{\alpha}}{c}\cdot\frac{2\dot{\alpha} + 4\dot{\beta}}{c} = \frac{e^{2\alpha}\dot{\alpha}(\dot{\alpha} + 2\dot{\beta})}{c^2}.$$

Combine:

$$R_{xx} = -\frac{e^{2\alpha}}{c^2}\big(\ddot{\alpha} + \dot{\alpha}^2 + 2\dot{\alpha}\dot{\beta}\big).$$

R_{yy} ($= R_{zz}$ by symmetry). Same procedure with $\beta \leftrightarrow \alpha$ in the relevant slots:

$$R_{yy} = -\frac{e^{2\beta}}{c^2}\big(\ddot{\beta} + \dot{\alpha}\dot{\beta} + 2\dot{\beta}^2\big).$$

Ricci scalar.

$$R = g^{00}R_{00} + e^{-2\alpha}R_{xx} + 2e^{-2\beta}R_{yy}.$$

Substitute:

$$R = -\frac{2}{c^2}\big(\ddot{\alpha} + 2\ddot{\beta} + \dot{\alpha}^2 + 3\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta}\big).$$

Einstein equations $G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}Rg_{\mu\nu} = -\tfrac{8\pi G}{c^4}T_{\mu\nu}$.
tt-component, $g_{00} = -1$, $T_{00} = \rho c^2$:

$$G_{00} = R_{00} + \tfrac{1}{2}R = -\frac{1}{c^2}\big(\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta}\big).$$

Set $G_{00} = -\tfrac{8\pi G}{c^4}\rho c^2$ and cancel $-1/c^2$:

$$\boxed{\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta} = 8\pi G\rho.} \tag{17}$$

xx-component, $g_{xx} = e^{2\alpha}$, $T_{xx} = g_{xx}T^x{}_x = e^{2\alpha}w_A\rho c^2$:

$$G_{xx} = R_{xx} - \tfrac{1}{2}R\,e^{2\alpha} = \frac{e^{2\alpha}}{c^2}\big(2\ddot{\beta} + 3\dot{\beta}^2\big).$$

Set $G_{xx} = -\frac{8\pi G}{c^4} e^{2\alpha} w_A \rho c^2$ and cancel $e^{2\alpha}/c^2$:

$$\boxed{2\ddot{\beta} + 3\dot{\beta}^2 = -8\pi G w_A \rho.} \tag{18}$$

yy-component, $T_{yy} = e^{2\beta} w_B \rho c^2$:

$$G_{yy} = \frac{e^{2\beta}}{c^2}\big(\ddot{\alpha} + \ddot{\beta} + \dot{\alpha}^2 + \dot{\beta}^2 + \dot{\alpha}\dot{\beta}\big).$$

Cancel $e^{2\beta}/c^2$:

$$\boxed{\ddot{\alpha} + \ddot{\beta} + \dot{\alpha}^2 + \dot{\beta}^2 + \dot{\alpha}\dot{\beta} = -8\pi G w_B \rho.} \tag{19}$$

(d) Cosmological-constant case $w_A = w_B = -1$

From (16) the exponent is 0:

$$\rho = \rho_0 = \text{const.}$$

Search for fixed point $\dot{\alpha} = \dot{\beta} = H$, $\ddot{\alpha} = \ddot{\beta} = 0$. Equation (17):

$$3H^2 = 8\pi G\rho_0 \implies H = \sqrt{\tfrac{8\pi G\rho_0}{3}} \equiv q.$$

Check (18): $0 + 3H^2 = 8\pi G\rho_0$. (19): $0 + 0 + 3H^2 = 8\pi G\rho_0$. Both consistent. Hence isotropic exponential solution $\alpha = \beta = qt$.

Linear stability. Set $\alpha = qt + \delta\alpha$, $\beta = qt + \delta\beta$, both small. Linearise (17):

$$2q\,\dot{\delta\beta} + 2q(\dot{\delta\alpha} + \dot{\delta\beta}) = 0.$$

Simplify:

$$\dot{\delta\alpha} + 2\dot{\delta\beta} = 0.$$

Linearise (18):

$$2\,\ddot{\delta\beta} + 6q\,\dot{\delta\beta} = 0.$$

Solve:

$$\dot{\delta\beta} \propto e^{-3qt} \xrightarrow{t\to\infty} 0.$$

And $\dot{\delta\alpha} = -2\dot{\delta\beta} \to 0$. Both shears decay; the isotropic de Sitter solution is stable.

$$\boxed{\rho = \text{const}, \quad \alpha = \beta = qt, \quad \text{shear} \propto e^{-3qt}.}$$

(e) Physical volume of a comoving region

Volume element of a constant-t slice:

$$\mathrm{d}V = \sqrt{|g_3|}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = e^{\alpha+2\beta}\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z.$$

Insert $\alpha = \beta = qt$ from (d):

$$V(t) = V_0\,e^{3qt}.$$

$$\boxed{V(t) \propto e^{3qt}.} \tag{20}$$

(f) Detecting cosmic anisotropy

Two probes are:

(i) CMB anisotropy pattern. Bianchi-I expansion imprints a direction-dependent quadrupole / octopole in the CMB temperature map (different recession rates in x vs y, z produce a preferred-axis signature). Stack with foreground-cleaned all-sky maps (Planck) and constrain shear $\Sigma \equiv (\dot{\alpha} - \dot{\beta})/H$ from the absence of an excess large-angle quadrupole.

(ii) Direction-dependent Hubble diagram. Compile Type Ia supernovae (or BAO standard rulers, or strong-lensing time delays) across the sky and fit $H_0(\hat{n})$ as a function of direction. A non-zero dipole / quadrupole in $H_0(\hat{n})$ above the peculiar-velocity floor implies anisotropic expansion.

Question 4 — Gravitational waves from a radial binary

Problem. Two equal masses μ on the z-axis, separation $\Delta z(t) = L + a\cos\omega t$. (a) Identify configuration; compute $I^{ij} = \int T^{00} r^i r^j \mathrm{d}^3 r$. (b) Linearised gravity, find $\bar{h}_{ij}^{TT}$ propagating along $\hat{x}$. (c) Motion of a small ring of test masses centred on, perpendicular to, $\hat{x}$. (d) Total GW luminosity. (e) Energy radiated per period; discuss L/a dependence.

(a) Physical configuration and quadrupole moment

A pair of equal masses on a spring along $\hat{z}$ (or two stars in a radial-only oscillation about a fixed mean separation L, amplitude a). Place the centre of mass at the origin:

$$z_{1,2}(t) = \mp\tfrac{1}{2}\big(L + a\cos\omega t\big), \qquad x_{1,2} = y_{1,2} = 0.$$

For point particles, $T^{00}(t,\vec{r}) = \mu c^2\big[\delta^3(\vec{r}-\vec{r}_1) + \delta^3(\vec{r}-\vec{r}_2)\big]$. Insert into the integral:

$$I^{ij} = \mu c^2\big(r_1^i r_1^j + r_2^i r_2^j\big).$$

Only $i = j = z$ is non-zero, with $z_1^2 = z_2^2 = (L + a\cos\omega t)^2/4$:

$$I^{zz} = 2\mu c^2 \cdot \frac{(L + a\cos\omega t)^2}{4} = \frac{\mu c^2}{2}\big(L + a\cos\omega t\big)^2.$$

$$\boxed{I^{ij} = \delta_z^i \delta_z^j \, \frac{\mu c^2}{2}\big(L + a\cos\omega t\big)^2.} \tag{21}$$

(b) Transverse-traceless metric perturbation along $\hat{x}$

Linearised quadrupole formula (Lorenz gauge, far zone):

$$\bar{h}^{ij}(t,\vec{r}) = \frac{2G}{c^4 r}\,\ddot{I}^{ij}\big(t - r/c\big). \tag{22}$$

Differentiate I^{zz} once:

$$\dot{I}^{zz} = \mu c^2 (L + a\cos\omega t)(-a\omega\sin\omega t).$$

Differentiate again:

$$\ddot{I}^{zz} = \mu c^2\big[a^2\omega^2\sin^2\omega t - (L + a\cos\omega t)a\omega^2\cos\omega t\big].$$

Use $\sin^2\omega t = \frac{1}{2}(1 - \cos 2\omega t)$ and $\cos^2\omega t = \frac{1}{2}(1 + \cos 2\omega t)$:

$$\ddot{I}^{zz} = -\mu c^2\omega^2\big[La\cos\omega t + a^2\cos 2\omega t\big].$$

Project to TT along $\hat{n} = \hat{x}$. Then $P_{ij} = \mathrm{diag}(0,1,1)$ in (x,y,z), and for a tensor with only A_{zz} nonzero:

$$A^{TT}_{ij} = P_{iz}P_{jz}A_{zz} - \tfrac{1}{2}P_{ij}A_{zz}.$$

Read off the non-vanishing components:

$$A^{TT}_{zz} = \big(1 - \tfrac{1}{2}\big)A_{zz} = \tfrac{1}{2}A_{zz}, \qquad A^{TT}_{yy} = -\tfrac{1}{2}A_{zz}, \qquad A^{TT}_{yz} = 0.$$

Apply to $\bar{h}^{ij}$ via (22) with $u \equiv t - r/c$:

$$\boxed{\bar{h}^{TT}_{zz} = -\bar{h}^{TT}_{yy} = -\frac{G\mu\omega^2}{c^2 r}\big[La\cos\omega u + a^2\cos 2\omega u\big].} \tag{23}$$

(c) Motion of the test-mass ring

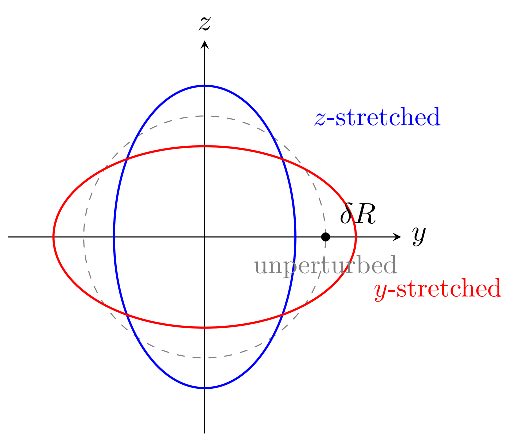
The wave is $+$-polarised along (y,z). Geodesic-deviation gives $\ddot{\xi}^i = \frac{1}{2}\ddot{\bar{h}}^{TT}_{ij}\xi^j$, integrated:

$$\xi^i(t) = \xi^i_0 + \tfrac{1}{2}\bar{h}^{TT}_{ij}(t)\xi^j_0.$$

Initial ring of radius δR in the (y,z) plane:

$$\delta y(t) = \delta R\cos\varphi\big[1 - \tfrac{1}{2}\bar{h}^{TT}_{zz}(t)\big], \qquad \delta z(t) = \delta R\sin\varphi\big[1 + \tfrac{1}{2}\bar{h}^{TT}_{zz}(t)\big].$$

Hence the circle deforms into an ellipse whose principal axes coincide with $\hat{y},\hat{z}$ and whose strain alternates sign at the two harmonics ω (dominant when $L \gg a$) and 2ω. The ring breathes between a z-stretched / y-squeezed ellipse and a y-stretched / z-squeezed ellipse, with constant area to linear order in $\bar{h}$.



(d) Total gravitational-wave luminosity

Use the standard Einstein quadrupole formula in terms of the trace-free reduced quadrupole $\bar{Q}_{ij} = Q_{ij} - \frac{1}{3}Q^k{}_k\delta_{ij}$ with $Q_{ij} \equiv I_{ij}/c^2$:

$$\mathcal{L} = \frac{G}{5c^5}\big\langle \dddot{\bar{Q}}_{ij}\dddot{\bar{Q}}^{ij}\big\rangle. \tag{24}$$

Read $Q_{zz} = \mu(L + a\cos\omega t)^2/2$, all other $Q_{ij} = 0$. Trace $Q^k{}_k = Q_{zz}$:

$$\bar{Q}_{zz} = \tfrac{2}{3}Q_{zz}, \qquad \bar{Q}_{xx} = \bar{Q}_{yy} = -\tfrac{1}{3}Q_{zz}.$$

Differentiate Q_{zz} three times. Expand:

$$Q_{zz} = \tfrac{\mu}{2}\big[L^2 + 2La\cos\omega t + \tfrac{a^2}{2} + \tfrac{a^2}{2}\cos 2\omega t\big].$$

Triple time derivative:

$$\dddot{Q}_{zz} = \mu\big[La\omega^3 \sin\omega t + 2a^2\omega^3 \sin 2\omega t\big] \equiv \mu S(t).$$

Square and time-average ($\langle \sin^2 \omega t\rangle = \langle \sin^2 2\omega t\rangle = \frac{1}{2}$, cross term averages to zero):

$$\langle S^2\rangle = \tfrac{1}{2}L^2a^2\omega^6 + 2a^4\omega^6 = \tfrac{1}{2}a^2\omega^6\big(L^2+4a^2\big).$$

Sum the trace-free components:

$$\dddot{Q}_{ij}\dddot{Q}^{ij} = \big(\tfrac{4}{9} + 2\cdot\tfrac{1}{9}\big)\mu^2 S^2 = \tfrac{2}{3}\mu^2 S^2.$$

Average:

$$\langle \dddot{Q}_{ij}\dddot{Q}^{ij}\rangle = \tfrac{\mu^2 a^2\omega^6(L^2+4a^2)}{3}.$$

Insert into (24):

$$\boxed{\mathcal{L} = \frac{G\mu^2\omega^6 a^2\big(L^2+4a^2\big)}{15\,c^5}.} \tag{25}$$

(e) Energy radiated per period

Period $T = 2\pi/\omega$:

$$\Delta E = \mathcal{L}\,T = \frac{2\pi G\mu^2\omega^5 a^2(L^2+4a^2)}{15c^5}.$$

Maximum speed of each mass: $\dot{z}_{1,2} = \pm\frac{1}{2}a\omega\sin\omega t$, so $v_{\max} = a\omega/2$, giving $a^2\omega^2 = 4v_{\max}^2$. Define a translational characteristic velocity $v_L \equiv L\omega/2$ (max speed of a hypothetical mass moving with relative-equilibrium phase $L\omega/2$), so $L^2\omega^2 = 4v_L^2$. Substitute:

$$\Delta E = \frac{2\pi G\mu^2\omega^3}{15c^5}\big(L^2+4a^2\big)\cdot 4v_{\max}^2 = \frac{8\pi G\mu^2\omega^3}{15c^5}\big(L^2+4a^2\big)v_{\max}^2.$$

Or, eliminating L, a entirely:

$$\boxed{\Delta E = \frac{8\pi G\mu^2\omega}{15c^5}\big(4v_L^2 v_{\max}^2 + 4v_{\max}^4\big) = \frac{32\pi G\mu^2\omega\, v_{\max}^2}{15c^5}\big(v_L^2 + v_{\max}^2\big).} \tag{26}$$

Discussion of L/a.

- $L/a \gg 1$ (small radial oscillation about a large separation): the ω-harmonic dominates, $\Delta E \approx \frac{2\pi G\mu^2\omega^3 L^2 a^2}{15c^5} \propto (L/a)^2 a^4\omega^5$. Energy loss grows as $(L/a)^2$, so the radiation is sourced by the mass-monopole modulation and scales linearly with the equilibrium length squared.

- $L/a \to 0$ (pure breathing through the centre): only the 2ω harmonic survives, $\Delta E \to \frac{8\pi G\mu^2\omega^3 a^4}{15c^5} = \frac{32\pi G\mu^2\omega v_{\max}^4}{15c^5}$. Independent of L; this is the lowest-order quadrupole emission of a symmetric oscillator.

- Crossover at $L/a \sim 2$: the two contributions are comparable, and the GW signal contains a clean two-tone spectrum at ω and 2ω.